

Academic Department Portal

Course Information

Course: CS5450 Mobile Programming
Challenge: React Native Mobile Mini Portal
Group: #5 Academic Department Portal

Project Description

The Academic Department Portal is a React Native mobile application developed using Expo Router and Firebase. The application provides students with access to department resources, course information, faculty contacts, announcements, and academic calendar information through a centralized portal.

The portal includes secure user authentication using Firebase Authentication and stores user profile information in Firebase Firestore.

Features

User Authentication

- User Registration
- User Login
- User Logout
- Firebase Authentication

Personalized Dashboard

- Displays the logged-in user's name
- Card-based dashboard navigation

Academic Services

- Course Listings
- Faculty Directory
- Student Resources
- Department Announcements
- Academic Calendar

Search Functionality

- Search courses
- Search faculty
- Search resources

Database Integration

- Firebase Firestore
 - User profile storage
-

Technologies Used

- React Native
 - Expo Router
 - TypeScript
 - Firebase Authentication
 - Firebase Firestore
 - Android Emulator / Android Device
-

Project Structure

None

src/

└─ app/

├─ index.tsx

├─ login.tsx

├─ register.tsx

├─ dashboard.tsx

├─ courses.tsx

├─ faculty.tsx

├─ resources.tsx

├─ search.tsx

├─ announcements.tsx

├─ calendar.tsx

└─ _layout.tsx

[firebase.js](#)

Firestore Configuration

Create a Firestore project using Firestore Console. Enable:

1. Authentication
 - Email/Password Sign-In
2. Firestore Database

Update the firebase.js file with your Firestore configuration settings.

Installation Instructions

Step 1

Install Node.js.

Step 2

Install project dependencies:

```
None  
npm install
```

Step 3

Start the Expo development server:

```
None  
npx expo start
```

Step 4

Run the application:

- Press "a" to launch Android Emulator
 - OR click the localhost link for Web deployment
 - OR scan the QR code using Expo Go on Android
-

Testing

Test the following functionality:

- Register a new account
 - Login with existing account
 - View dashboard
 - Open Courses page
 - Open Faculty Directory page
 - Open Resources page
 - Open Search page
 - Open Announcements page
 - Open Academic Calendar page
 - Logout successfully
-

Future Improvements

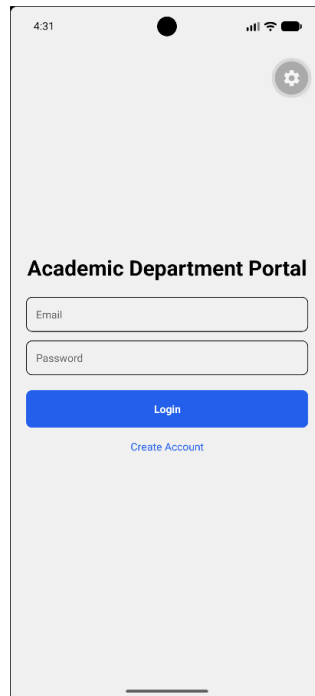
- Profile Management
 - Dynamic Course Database
 - Dynamic Faculty Database
 - Push Notifications
 - Admin Portal
 - Dark Mode Support
-

Github Repository

Repository Link: <https://github.com/unguyen11/AcademicDepartmentPortal>

Screenshots

1. Login Screen



A screenshot of a mobile application's login screen. The status bar at the top shows the time 4:31, signal strength, and battery level. A settings gear icon is in the top right corner. The main heading is "Academic Department Portal". Below it are two input fields labeled "Email" and "Password". A blue "Login" button is positioned below the password field, with a "Create Account" link underneath it.

4:31

Academic Department Portal

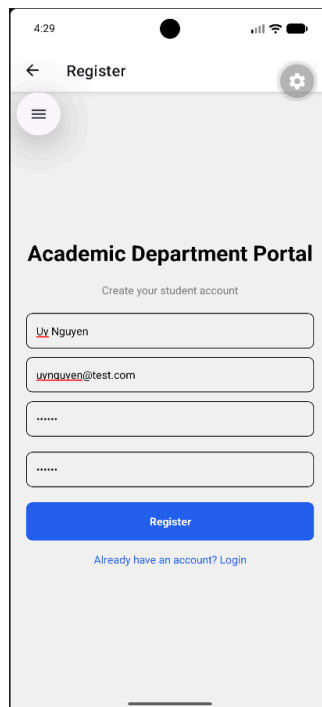
Email

Password

Login

Create Account

2. Registration Screen



A screenshot of a mobile application's registration screen. The status bar at the top shows the time 4:29, signal strength, and battery level. A back arrow and the word "Register" are in the top left, and a settings gear icon is in the top right. A hamburger menu icon is on the left. The main heading is "Academic Department Portal", followed by the subtext "Create your student account". There are four input fields: the first contains "Uy Nguyen", the second contains "uynquyen@test.com", and the next two are masked with asterisks. A blue "Register" button is below the fields, with a link "Already have an account? Login" underneath it.

4:29

← Register

Academic Department Portal

Create your student account

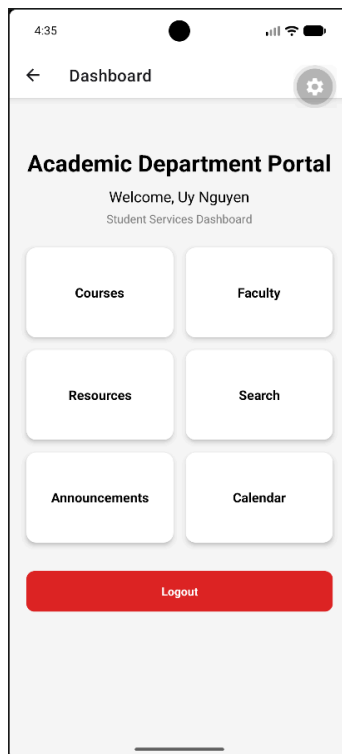
Uy Nguyen

uynquyen@test.com

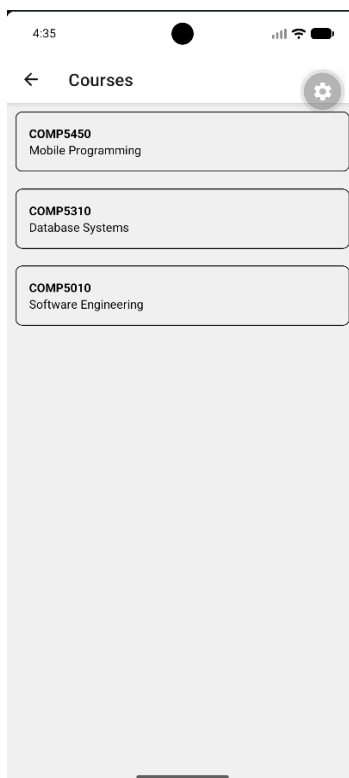
Register

Already have an account? Login

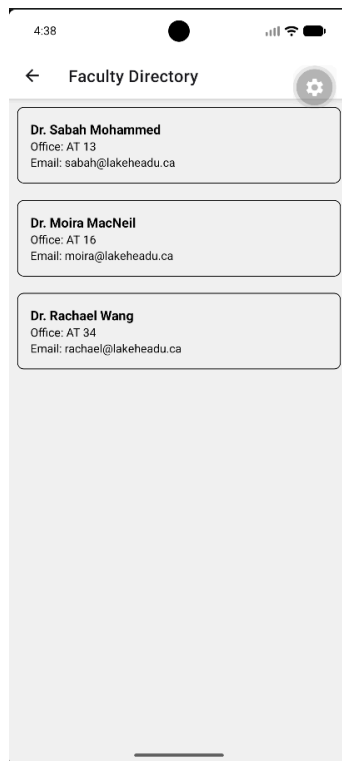
3. Dashboard Screen



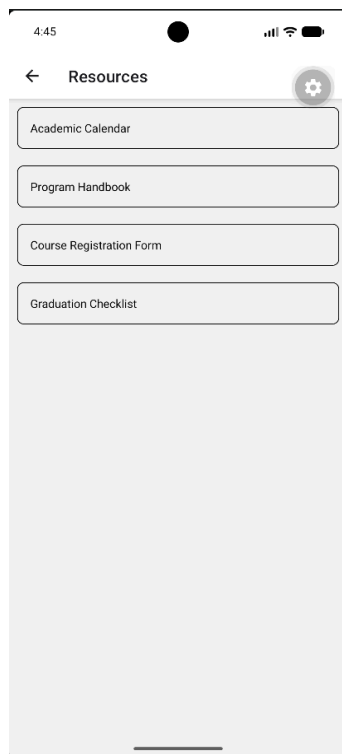
4. Courses Screen



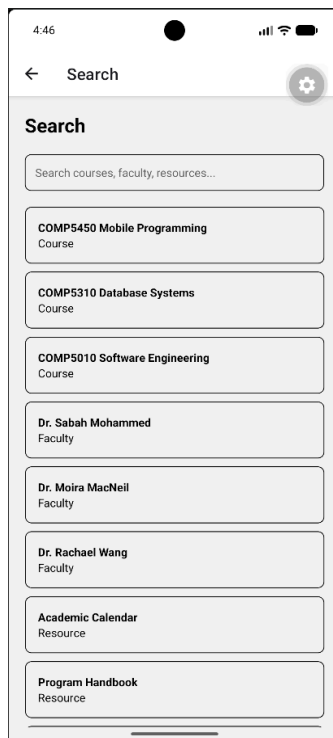
5. Faculty Directory Screen



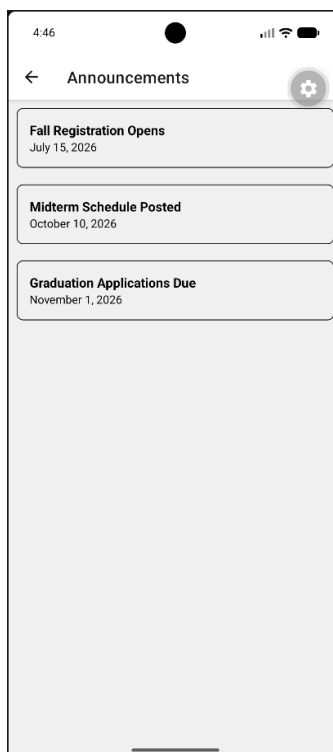
6. Resources Screen



7. Search Screen



8. Announcements Screen



9. Academic Calendar Screen

